Mining production report

Weyland-Yutani Corporation

Google Sheets, Data!A:C

Period: 2099-11-02 to 2099-12-11. Generated 2026-11-03 17:50 UTC.

Analysis settings

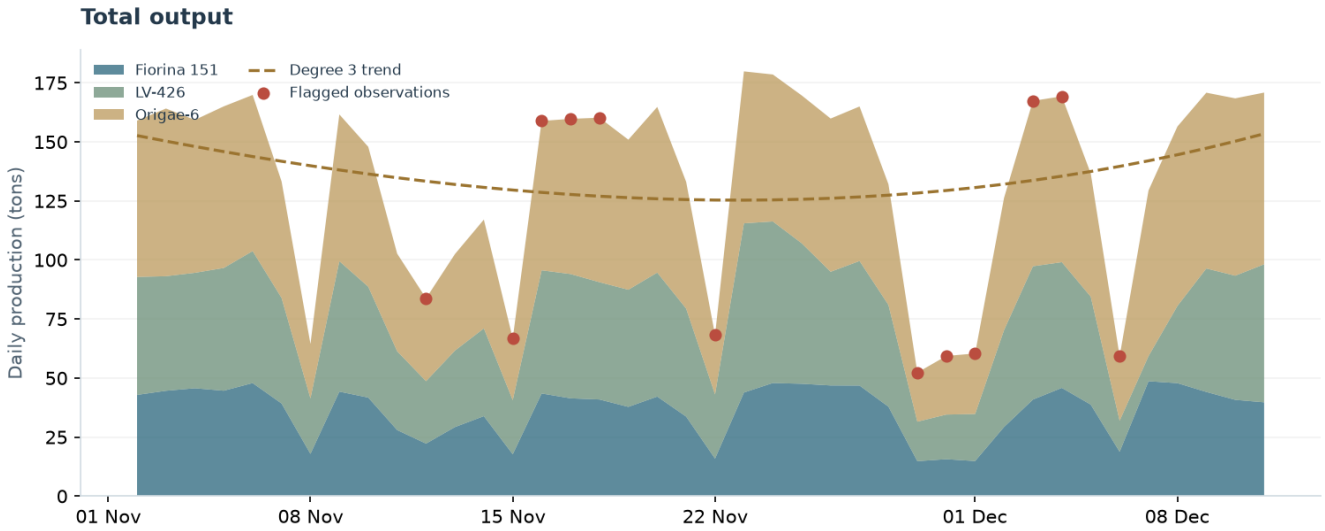
IQR multiplier 1.5; z-score threshold 3; moving-average window 7 previous observations; distance threshold 30%; Grubbs alpha 0.05. Charts: stacked, polynomial degree 3, highlight: Any test.

Standard deviation uses n-1. IQR uses linearly interpolated quartiles. The moving average excludes the current observation and needs a full window. Spike/drop direction compares output with that preceding mean, or the series median when the window is unavailable. Grubbs is a two-sided single-outlier test, at most one result per series. Its normality assumption can be violated by trends and weekly patterns; flags are candidates for review.

Production statistics

Series	Days	Mean	Std. dev.	Median	IQR
Total output	40	135.13	40.55	157.69	51.51
Fiorina 151	40	36.65	11.11	40.90	15.36
LV-426	40	43.06	15.07	48.33	19.96
Origae-6	40	55.42	17.21	62.92	23.84

Total output



Flag counts - IQR: 0, Z-score: 0, Moving average: 12, Grubbs: 0. Distinct observations: 12.
IQR acceptance interval: 36.22 to 242.27 tons.
Grubbs: G = 2.0426; critical G = 3.0361.

Anomaly details

2099-11-12: drop at 83.75 tons

Tests: Moving average. Reference output: 134.97 tons. Absolute z-score: 1.27. Distance from preceding moving average: 37.9%.

2099-11-15: drop at 66.75 tons

Tests: Moving average. Reference output: 111.42 tons. Absolute z-score: 1.69. Distance from preceding moving average: 40.1%.

2099-11-16: spike at 158.87 tons

Tests: Moving average. Reference output: 111.76 tons. Absolute z-score: 0.59. Distance from preceding moving average: 42.2%.

2099-11-17: spike at 159.75 tons

Tests: Moving average. Reference output: 111.36 tons. Absolute z-score: 0.61. Distance from preceding moving average: 43.4%.

2099-11-18: spike at 160.24 tons

Tests: Moving average. Reference output: 113.06 tons. Absolute z-score: 0.62. Distance from preceding moving average: 41.7%.

2099-11-22: drop at 68.24 tons

Tests: Moving average. Reference output: 142.06 tons. Absolute z-score: 1.65. Distance from preceding moving average: 52.0%.

2099-11-29: drop at 52.31 tons

Tests: Moving average. Reference output: 150.46 tons. Absolute z-score: 2.04. Distance from preceding moving average: 65.2%.

2099-11-30: drop at 59.33 tons

Tests: Moving average. Reference output: 148.18 tons. Absolute z-score: 1.87. Distance from preceding moving average: 60.0%.

2099-12-01: drop at 60.41 tons

Tests: Moving average. Reference output: 130.97 tons. Absolute z-score: 1.84. Distance from preceding moving average: 53.9%.

2099-12-03: spike at 167.36 tons

Tests: Moving average. Reference output: 107.89 tons. Absolute z-score: 0.79. Distance from preceding moving average: 55.1%.

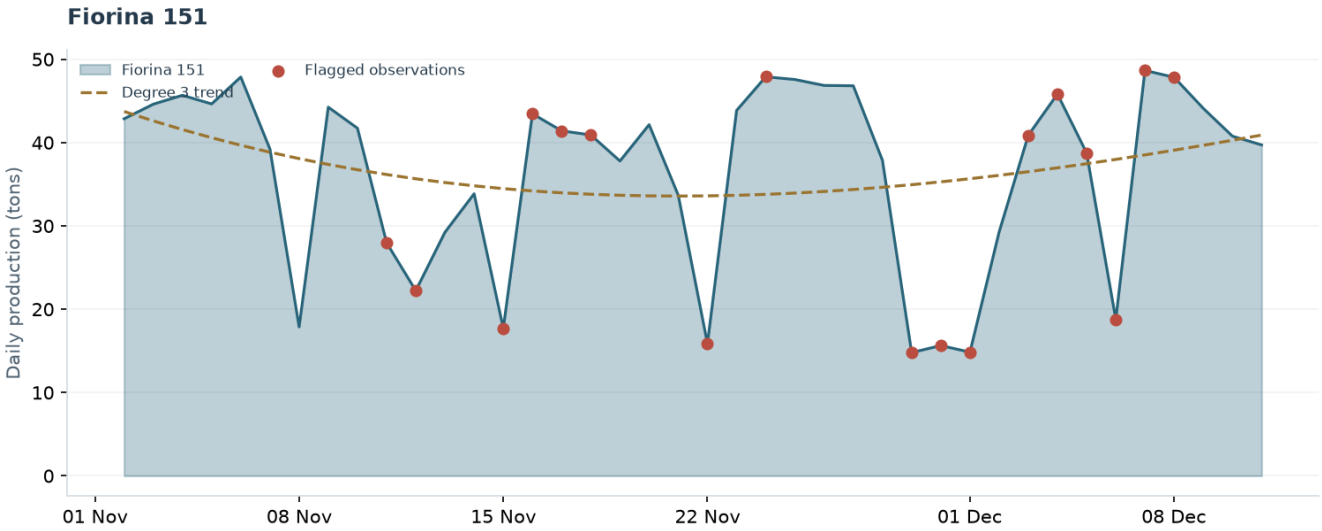
2099-12-04: spike at 169.30 tons

Tests: Moving average. Reference output: 108.96 tons. Absolute z-score: 0.84. Distance from preceding moving average: 55.4%.

2099-12-06: drop at 59.21 tons

Tests: Moving average. Reference output: 110.23 tons. Absolute z-score: 1.87. Distance from preceding moving average: 46.3%.

Fiorina 151



Flag counts - IQR: 0, Z-score: 0, Moving average: 17, Grubbs: 0. Distinct observations: 17.
IQR acceptance interval: 6.23 to 67.65 tons.
Grubbs: G = 1.9665; critical G = 3.0361.

Anomaly details

2099-11-11: drop at 27.97 tons

Tests: Moving average. Reference output: 40.18 tons. Absolute z-score: 0.78. Distance from preceding moving average: 30.4%.

2099-11-12: drop at 22.23 tons

Tests: Moving average. Reference output: 37.65 tons. Absolute z-score: 1.30. Distance from preceding moving average: 41.0%.

2099-11-15: drop at 17.71 tons

Tests: Moving average. Reference output: 31.02 tons. Absolute z-score: 1.70. Distance from preceding moving average: 42.9%.

2099-11-16: spike at 43.46 tons

Tests: Moving average. Reference output: 31.00 tons. Absolute z-score: 0.61. Distance from preceding moving average: 40.2%.

2099-11-17: spike at 41.41 tons

Tests: Moving average. Reference output: 30.89 tons. Absolute z-score: 0.43. Distance from preceding moving average: 34.1%.

2099-11-18: spike at 40.92 tons

Tests: Moving average. Reference output: 30.84 tons. Absolute z-score: 0.38. Distance from preceding moving average: 32.7%.

2099-11-22: drop at 15.87 tons

Tests: Moving average. Reference output: 36.73 tons. Absolute z-score: 1.87. Distance from preceding moving average: 56.8%.

2099-11-24: spike at 47.90 tons

Tests: Moving average. Reference output: 36.52 tons. Absolute z-score: 1.01. Distance from preceding moving average: 31.1%.

2099-11-29: drop at 14.80 tons

Tests: Moving average. Reference output: 40.96 tons. Absolute z-score: 1.97. Distance from preceding moving average: 63.9%.

2099-11-30: drop at 15.64 tons

Tests: Moving average. Reference output: 40.81 tons. Absolute z-score: 1.89. Distance from preceding moving average: 61.7%.

2099-12-01: drop at 14.85 tons

Tests: Moving average. Reference output: 36.78 tons. Absolute z-score: 1.96. Distance from preceding moving average: 59.6%.

2099-12-03: spike at 40.88 tons

Tests: Moving average. Reference output: 29.44 tons. Absolute z-score: 0.38. Distance from preceding moving average: 38.8%.

2099-12-04: spike at 45.83 tons

Tests: Moving average. Reference output: 28.59 tons. Absolute z-score: 0.83. Distance from preceding moving average: 60.3%.

2099-12-05: spike at 38.75 tons

Tests: Moving average. Reference output: 28.45 tons. Absolute z-score: 0.19. Distance from preceding moving average: 36.2%.

2099-12-06: drop at 18.78 tons

Tests: Moving average. Reference output: 28.57 tons. Absolute z-score: 1.61. Distance from preceding moving average: 34.3%.

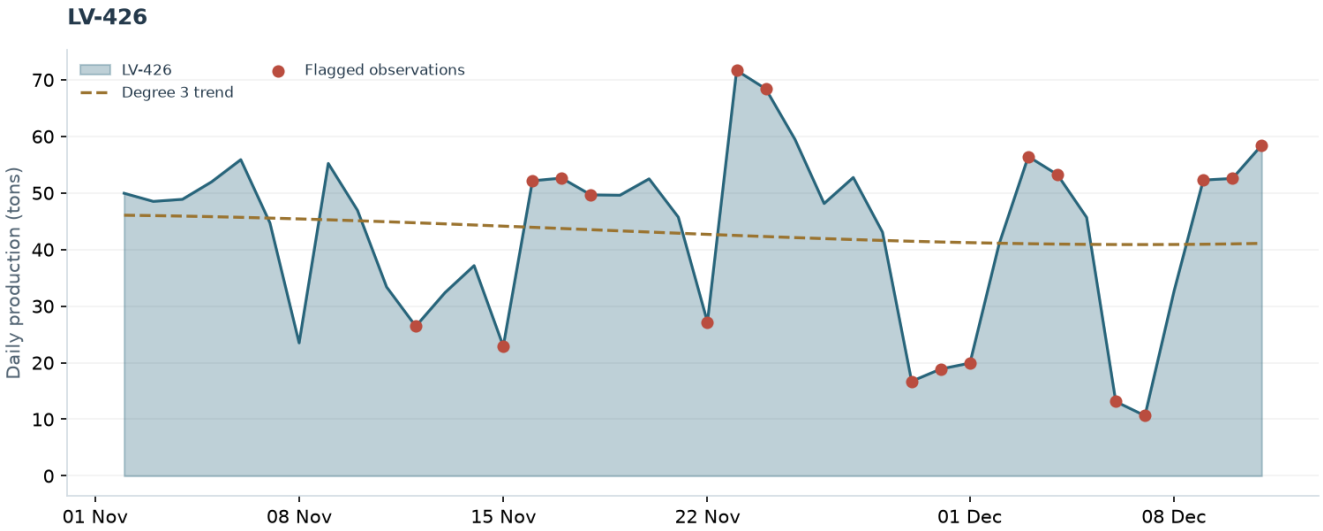
2099-12-07: spike at 48.66 tons

Tests: Moving average. Reference output: 29.14 tons. Absolute z-score: 1.08. Distance from preceding moving average: 67.0%.

2099-12-08: spike at 47.84 tons

Tests: Moving average. Reference output: 33.86 tons. Absolute z-score: 1.01. Distance from preceding moving average: 41.3%.

LV-426



Flag counts - IQR: 0, Z-score: 0, Moving average: 18, Grubbs: 0. Distinct observations: 18.
IQR acceptance interval: 2.67 to 82.53 tons.
Grubbs: G = 2.1518; critical G = 3.0361.

Anomaly details

2099-11-12: drop at 26.44 tons

Tests: Moving average. Reference output: 44.52 tons. Absolute z-score: 1.10. Distance from preceding moving average: 40.6%.

2099-11-15: drop at 22.92 tons

Tests: Moving average. Reference output: 36.43 tons. Absolute z-score: 1.34. Distance from preceding moving average: 37.1%.

2099-11-16: spike at 52.16 tons

Tests: Moving average. Reference output: 36.35 tons. Absolute z-score: 0.60. Distance from preceding moving average: 43.5%.

2099-11-17: spike at 52.63 tons

Tests: Moving average. Reference output: 35.91 tons. Absolute z-score: 0.63. Distance from preceding moving average: 46.5%.

2099-11-18: spike at 49.67 tons

Tests: Moving average. Reference output: 36.72 tons. Absolute z-score: 0.44. Distance from preceding moving average: 35.3%.

2099-11-22: drop at 27.19 tons

Tests: Moving average. Reference output: 46.46 tons. Absolute z-score: 1.05. Distance from preceding moving average: 41.5%.

2099-11-23: spike at 71.66 tons

Tests: Moving average. Reference output: 47.07 tons. Absolute z-score: 1.90. Distance from preceding moving average: 52.2%.

2099-11-24: spike at 68.40 tons

Tests: Moving average. Reference output: 49.86 tons. Absolute z-score: 1.68. Distance from preceding moving average: 37.2%.

2099-11-29: drop at 16.72 tons

Tests: Moving average. Reference output: 52.96 tons. Absolute z-score: 1.75. Distance from preceding moving average: 68.4%.

2099-11-30: drop at 18.89 tons

Tests: Moving average. Reference output: 51.46 tons. Absolute z-score: 1.60. Distance from preceding moving average: 63.3%.

2099-12-01: drop at 19.93 tons

Tests: Moving average. Reference output: 43.92 tons. Absolute z-score: 1.53. Distance from preceding moving average: 54.6%.

2099-12-03: spike at 56.45 tons

Tests: Moving average. Reference output: 34.36 tons. Absolute z-score: 0.89. Distance from preceding moving average: 64.3%.

2099-12-04: spike at 53.24 tons

Tests: Moving average. Reference output: 35.55 tons. Absolute z-score: 0.68. Distance from preceding moving average: 49.8%.

2099-12-06: drop at 13.13 tons

Tests: Moving average. Reference output: 35.99 tons. Absolute z-score: 1.99. Distance from preceding moving average: 63.5%.

2099-12-07: drop at 10.63 tons

Tests: Moving average. Reference output: 35.48 tons. Absolute z-score: 2.15. Distance from preceding moving average: 70.0%.

2099-12-09: spike at 52.28 tons

Tests: Moving average. Reference output: 36.12 tons. Absolute z-score: 0.61. Distance from preceding moving average: 44.7%.

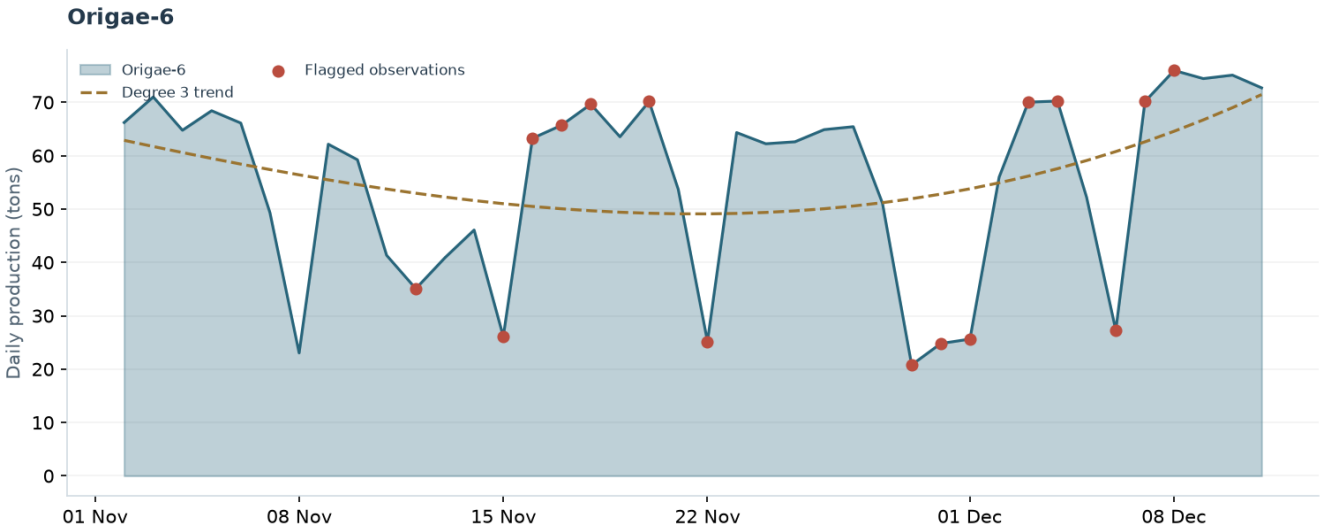
2099-12-10: spike at 52.57 tons

Tests: Moving average. Reference output: 37.73 tons. Absolute z-score: 0.63. Distance from preceding moving average: 39.3%.

2099-12-11: spike at 58.44 tons

Tests: Moving average. Reference output: 37.18 tons. Absolute z-score: 1.02. Distance from preceding moving average: 57.2%.

Origae-6



Flag counts - IQR: 0, Z-score: 0, Moving average: 15, Grubbs: 0. Distinct observations: 15.
IQR acceptance interval: 9.12 to 104.49 tons.
Grubbs: G = 2.0125; critical G = 3.0361.

Anomaly details

2099-11-12: drop at 35.08 tons

Tests: Moving average. Reference output: 52.80 tons. Absolute z-score: 1.18. Distance from preceding moving average: 33.6%.

2099-11-15: drop at 26.12 tons

Tests: Moving average. Reference output: 43.97 tons. Absolute z-score: 1.70. Distance from preceding moving average: 40.6%.

2099-11-16: spike at 63.25 tons

Tests: Moving average. Reference output: 44.41 tons. Absolute z-score: 0.46. Distance from preceding moving average: 42.4%.

2099-11-17: spike at 65.71 tons

Tests: Moving average. Reference output: 44.57 tons. Absolute z-score: 0.60. Distance from preceding moving average: 47.4%.

2099-11-18: spike at 69.65 tons

Tests: Moving average. Reference output: 45.49 tons. Absolute z-score: 0.83. Distance from preceding moving average: 53.1%.

2099-11-20: spike at 70.15 tons

Tests: Moving average. Reference output: 53.60 tons. Absolute z-score: 0.86. Distance from preceding moving average: 30.9%.

2099-11-22: drop at 25.18 tons

Tests: Moving average. Reference output: 58.87 tons. Absolute z-score: 1.76. Distance from preceding moving average: 57.2%.

2099-11-29: drop at 20.79 tons

Tests: Moving average. Reference output: 56.54 tons. Absolute z-score: 2.01. Distance from preceding moving average: 63.2%.

2099-11-30: drop at 24.80 tons

Tests: Moving average. Reference output: 55.91 tons. Absolute z-score: 1.78. Distance from preceding moving average: 55.6%.

2099-12-01: drop at 25.63 tons

Tests: Moving average. Reference output: 50.26 tons. Absolute z-score: 1.73. Distance from preceding moving average: 49.0%.

2099-12-03: spike at 70.03 tons

Tests: Moving average. Reference output: 44.09 tons. Absolute z-score: 0.85. Distance from preceding moving average: 58.8%.

2099-12-04: spike at 70.23 tons

Tests: Moving average. Reference output: 44.82 tons. Absolute z-score: 0.86. Distance from preceding moving average: 56.7%.

2099-12-06: drop at 27.30 tons

Tests: Moving average. Reference output: 45.66 tons. Absolute z-score: 1.63. Distance from preceding moving average: 40.2%.

2099-12-07: spike at 70.14 tons

Tests: Moving average. Reference output: 46.59 tons. Absolute z-score: 0.86. Distance from preceding moving average: 50.5%.

2099-12-08: spike at 75.97 tons

Tests: Moving average. Reference output: 53.07 tons. Absolute z-score: 1.19. Distance from preceding moving average: 43.2%.